

adapol: Adaptive pole-fitting for quantum many-body physics

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Software

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Summary

The Green's function approach to quantum many-body physics aims to replace high-dimensional wavefunctions with correlation functions which are more closely related to experimental observables of interest, such as spectral and response functions. Finite temperature quantities, such as the Green's function, self-energy, and hybridization functions, are often represented in the discrete "Matsubara" domain on the imaginary frequency axis (Matsubara, 1955). A variety of physical observables can be directly recovered from the Matsubara Green's function, and many quantities of interest can be calculated more efficiently in this formalism.

A common computational task within this framework is decomposing a Matsubara function into a sum of simple poles:

$$G(i\nu_n) \approx \sum_{k=1}^M \frac{R_k}{i\nu_n - p_k}.$$

Here, $G(i\nu_n)$ is in general an $m \times m$ matrix-valued function of the Matsubara frequency point $i\nu_n = (2n+1)\pi/\beta$ for fermionic functions, and $i\nu_n = 2n\pi/\beta$ for bosonic functions, with β representing the inverse temperature, $n \in \mathbb{Z}$, and m the number of quantum states or spin-orbitals. The p_k are real-valued pole locations, and the R_k are the corresponding matrix-valued residues. In applications such as hybridization fitting, the poles and residues define an effective non-interacting model, with the p_k playing the role of energy levels, and it is often desirable to obtain an accurate fit with as few poles as possible.

Since the pole locations enter the approximation nonlinearly and are shared by all components of a matrix-valued function, a best fit from Matsubara frequency data cannot be obtained component-wise, leading to a highly non-convex optimization landscape. *adapol* ("add a pole") is a Python package implementing the adaptive pole-fitting procedure outlined in (Huang et al., 2023, 2025). The method uses a modified version of the AAA rational approximation algorithm (Nakatsukasa et al., 2018) to obtain a guess of the pole locations p_k , which can optionally be refined by non-convex optimization. The residues R_k are then obtained by a linear least-squares fit. This procedure has been shown to provide an accurate and compact fit of Matsubara data in a black-box and noise-robust manner (Huang et al., 2025; Zima et al., 2026).

Statement of Need

The “pole-fitting” problem described above is a crucial step in various numerical methods, such as hybridization fitting for quantum impurity solvers (Georges et al., 1996) based on exact diagonalization (Caffarel & Krauth, 1994; Liebsch & Ishida, 2011; Mejuto-Zaera et al., 2020), perturbation theory (Huang et al., 2025; Kaye et al., 2024), and time evolution of matrix product states (Wolf et al., 2015; Zima et al., 2026), as well as certain approaches to analytic continuation of Matsubara Green’s functions (Fei, Yeh, & Gull, 2021; Fei, Yeh, Zgid, et al., 2021; Huang et al., 2023; Ying, 2022a, 2022b; Zhang et al., 2024; Zhang & Gull, 2024), and other perturbation theory-based diagrammatic methods (Gazizova et al., 2024, 2025). In many applications (e.g., dynamical mean-field theory), the pole-fitting step appears inside a self-consistent loop, requiring a black-box algorithm delivering results with controlled accuracy.

Although significant progress has been made in the past several years on developing algorithms to solve the pole-fitting problem (Huang et al., 2023, 2025; Mejuto-Zaera et al., 2020; Shinaoka & Nagai, 2021; Ying, 2022a, 2022b; Zhang et al., 2024; Zhang & Gull, 2024), a lack of widely-deployed and user-friendly software has limited the adoption of these methods. Practitioners often still rely on ad-hoc or older, easy-to-implement methods: for example, brute force optimization methods for pole-fitting, or Padé approximants for analytic continuation (Vidberg & Serene, 1977). `adapol` addresses this gap by providing a simple, self-contained interface with few user parameters, tailored for common applications.

State of the field

We note two primary approaches which have recently been pursued in the literature on the pole-fitting problem: methods based on (i) Prony’s method and its variants (Ying, 2022a, 2022b; Zhang et al., 2024; Zhang & Gull, 2024), and (ii) AAA rational approximation (Nakatsukasa et al., 2018) followed by non-convex optimization (Huang et al., 2023, 2025). The MiniPole Python package (Zhang et al., 2025) implements the Prony’s method-based approach described in (Zhang et al., 2024; Zhang & Gull, 2024), while `adapol` implements the AAA-based approach described in (Huang et al., 2023, 2025). These methods are distinct, and the availability of both packages will allow users to compare the two approaches.

Software design

`adapol` is a simple and self-contained package which can be incorporated into codes requiring Matsubara pole-fitting. Users can provide Matsubara data, or an existing pole expansion to be compressed; for example, a discrete Lehmann representation (DLR) (Kaye et al., 2022). They can choose to perform the fit with or without optimization-based post-processing of the AAA result, and can specify either a maximum number of poles or a target error tolerance. `adapol` functions are documented extensively both within the API reference documentation, and in example notebooks demonstrating various use cases and modes of operation. An interface to the TRIQS package (Parcollet et al., 2015) is also provided.

Research impact statement

In the context of analytic continuation, the AAA-based approach described in (Huang et al., 2023) is considered as one of the state-of-the-art methods, and has been cited extensively. For many other pole-fitting applications, reducing the number of poles required to achieve a given accuracy as much as possible is often crucial, as computational costs often scale exponentially with the number of poles. This is the case, for example, in exact diagonalization quantum impurity solvers (Caffarel & Krauth, 1994; Liebsch & Ishida, 2011; Mejuto-Zaera et al., 2020) and diagrammatic evaluation methods (Gazizova et al., 2024, 2025; Huang et

al., 2025; Kaye et al., 2024). Recent developments in tensor network-based quantum impurity solvers (Zima et al., 2026) also benefit substantially from compact pole approximations of the hybridization function, with a significant increase in computational cost observed as the size of this approximation grows. The adapol algorithm was shown in (Huang et al., 2025) to consistently yield a more compact pole approximation of a fixed, given Green's function than the generic DLR approach (Kaye et al., 2022), which is itself an exponential-in- β improvement over naive uniform frequency grid approaches.

AI usage disclosure

Generative AI tools such as Claude and Codex were used to assist in writing code, tests, documentation, and examples in the adapol package. The content produced by these tools was reviewed and edited by the authors.

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